

CropSage Database Design - Balanced Edition

A concise Supabase design that retains the complete per-table schema contract, deterministic scoring rules and security boundaries.

FINAL DECISION	Supabase Postgres with 11 application tables, PostGIS for farm geometry, one private provider-artifacts bucket, and Git for the crop catalog, Texas site manifest, scoring policy, schemas and source code.
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Persistence Flow

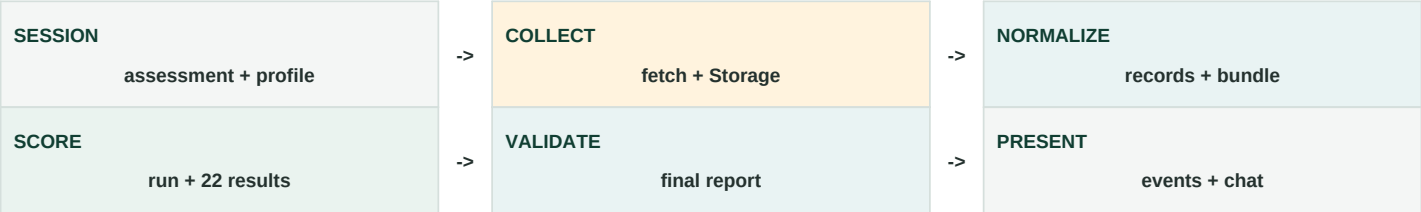


Table 1 - assessment_sessions

Column	Type / constraint	Purpose
id	uuid PK	Stable assessment and conversation identifier.
owner_user_id	uuid null FK auth.users	Optional signed-in owner; Auth is not one of the 11 application tables.
anonymous_token_hash	text null	Hash of the server-issued token; never store the raw token.
status	enum	active, completed, expired, archived.
active_profile_id	uuid null FK	Current immutable profile version.
active_run_id	uuid null FK	Last validated recommendation run.
structured_state_jsonb	jsonb	Pending slots, selected crop and safe UI references; not scoring truth.
created_at / updated_at	timestamptz	Lifecycle timestamps.
last_activity_at / expires_at	timestamptz	Retention and session expiry.

Sessions And Immutable Farm Profiles

Columns and rules required to retain anonymous/authenticated assessment state and versioned farmer inputs.

Table 2 - farm_profiles

Field group	Required columns and rules
Identity	id; session_id; external_profile_id; profile_version; input_hash; supersedes_profile_id; created_at. Unique(session_id, profile_id, profile_version).
Geometry	farm_point geography(Point,4326) NOT NULL; farm_boundary geography(Geometry,4326) NULL; location_hash; farm_name and location_label nullable.
Resolution	location_mode; representative_site_id nullable; resolved_region_id; region_assignment_method; region_resolver_version; timezone; timezone_resolution_method; location_source.
Planting	planned_date or planned_month, never both; flexibility_days; requested_crop_id nullable.
Irrigation	irrigation_jsonb nullable: availability, reliability, method, source, well/canal capacity and notes. Unknown remains unknown.
Overrides	soil_overrides_jsonb; current_soil_moisture_jsonb; recent_rainfall_jsonb; farmer_goal_jsonb.
Audit	source_metadata_jsonb; missing_fields_jsonb; profile_schema_version; completeness_notes.

TABLE 3 | PROVIDER REQUEST AND CACHE LIFECYCLE

Provider Fetches

A single request record captures provider submission, cache state, failures and the stable private artifact reference.

Table 3 - provider_fetches

Group	Columns	Rule
Identity	id; request_group_id; farm_profile_id nullable; representative_site_id nullable; location_hash	request_group_id links provider calls for one collection attempt.
Request	provider; endpoint; http_method; sanitized_parameters_jsonb; request_hash; cache_key	No secret headers, API keys or unsafe payload fragments.
FortyGuard	activity_id; poll_status; poll_count; filter_type; analytic_type; threshold_c; direction; granularity_m	Historical heatmap uses filter_type=4; exact parameters remain in JSONB.
Lifecycle	status; attempt_number; http_status; submitted_at; completed_at; fetched_at	status: queued, submitted, processing, completed, failed, timed_out, cancelled.
Cache	cache_state; cache_hit; expires_at; source_data_vintage	cache_state: live, fresh_cache, stale_cache, fallback, none.
Failure	error_code; safe_error_message; retryable	Errors are sanitized and never contain tokens or full sensitive responses.
Artifact	bucket_name; object_path; sha256; content_type; size_bytes; provider_timestamp; sanitization_status	Stable private object reference; never save a temporary signed URL.

Evidence Records And Immutable Bundles

Queryable normalized facts remain separate from the exact immutable EvidenceBundle snapshot supplied to scoring.

Table 4 - evidence_records

Group	Columns	Rule / meaning
Identity	id; provider_fetch_id; canonical_variable; dedupe_hash	One normalized fact or compact series per record.
Value	value_numeric; value_text; value_jsonb	CHECK an appropriate representation is populated; null means unknown, never zero.
Units	normalized_unit; source_field; source_unit; conversion_method	Canonical units follow EvidenceBundle 1.1.0.
Time	temporal_role; observed_from/to; valid_from/to; forecast_issued_at; source_timezone; normalized_timezone; aggregation_method	Separates climatology, history, recent, current and forecast.
Space	support_geometry; spatial_support; spatial_resolution_m	Point, farm area, tile or map unit.
Depth	depth_top_cm; depth_bottom_cm; depth_basis	Required for SSURGO horizons and soil-moisture layers.
Quality	quality_status; freshness_status; quality_flags_jsonb; warnings_jsonb	Statuses remain visible to scorer and validator.

Tables 5-6 - bundles and membership

Table	Required fields
evidence_bundles	id; external_bundle_id; farm_profile_id; schema_version; bundle_hash; location_hash; representative_site_id; region_manifest_version/hash; assembled_at; status; provider_coverage_jsonb; completeness_percent; freshness_summary_jsonb; invalid_numeric_count; bundle_jsonb; validation_jsonb.
evidence_bundle_records	evidence_bundle_id; evidence_record_id; inclusion_role; inclusion_order; required_for_scoring. Composite primary key(bundle_id, record_id).

Bundle Status Rules

Status	Meaning	Scoreable?
assembling	Collection or normalization incomplete.	No
validated	Contract, units, location and freshness checks passed.	Yes
partial	Optional evidence unavailable; missing fields explicit.	Only if essential-evidence policy passes
stale	Required artifact exceeds freshness policy.	Normally no; disclosed fallback only
invalid	Schema, coordinate, unit or numeric validation failed.	No
failed	Assembly produced no usable bundle.	No

REPRODUCTION RULE	Store the exact EvidenceBundle JSONB and SHA-256 hash. The scorer receives that immutable snapshot, not a fresh reconstruction from mutable evidence rows.
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Recommendation Runs And Crop Results

Every baseline and scenario execution retains its complete versioned input/output envelope and one auditable row per crop.

Table 7 - recommendation_runs

Group	Columns
Identity	id; session_id; farm_profile_id; evidence_bundle_id; requested_by_message_id nullable; created_at.
Mode	evaluation_mode; requested_crop_id nullable; status; baseline_or_scenario; parent_run_id nullable.
Scenario	scenario_batch_id; scenario_key; scenario_label; scenario_type; scenario_changes_jsonb; assumptions_jsonb.
Versions	profile version/hash; evidence schema version/hash; catalog version/hash; region manifest version/hash; engine version/Git commit; scoring policy version/hash; recommendation schema version.
Snapshots	engine_input_jsonb and engine_output_jsonb, both immutable.
Timing	started_at; completed_at; duration_ms; safe_error_code; safe_error_message.
Presentation	validated_presentation_jsonb nullable; only after render_allowed=true and never future scoring input.

Table 8 - crop_score_results

Group	Columns
Identity	id; recommendation_run_id; crop_id; crop_name. Unique(run_id, crop_id).
Outcome	status; eligibility; hard_filter_reasons_jsonb; rank; suitability_score; recommendation_band.
Confidence	confidence_score; confidence_band; evidence_coverage_percent; confidence_inputs_jsonb.
Calculation	category_components_jsonb; factors_jsonb; comparison_metrics_jsonb; applied_caps_jsonb.
Explanation inputs	strengths_jsonb; risks_jsonb; warnings_jsonb; reason_codes_jsonb.
Provenance	evidence_record_ids_jsonb; missing_evidence_jsonb; supplementary_evidence_jsonb.
Scenario	baseline_score; score_delta; baseline_rank; rank_delta.

Scenario Rules

Scenario	Saved change
in_season_baseline	No change; parent_run_id is null.
out_of_season	Changed planned date/month.
no_irrigation	Irrigation availability set to no.
reliable_irrigation	Availability yes; reliability reliable.
missing_or_lab_override	Hide optional evidence or supply laboratory soil override.

Render Authorization And Run Events

Validation remains deterministic; progress events expose status without prompts, secrets or hidden reasoning.

Table 9 - validation_reports

Column	Type / content
id / recommendation_run_id	uuid; unique final report per run.
validator_version	Version or Git commit of deterministic validator.
status	passed, passed_with_warnings, failed.
schema_checks_jsonb	Input/output schema results.
reconciliation_jsonb	Weights, totals, ranks, caps and 22-crop checks.
grounding_jsonb	Catalog crop IDs, thresholds, regions and evidence links.
freshness_jsonb	Provider coverage and expiry assessment.
errors_jsonb / warnings_jsonb	Structured machine-safe diagnostics.
render_allowed	boolean NOT NULL; true only after all blocking checks pass.
validated_at	timestampz.

Table 10 - run_events

Column	Purpose
id; session_id; run_id; provider_fetch_id	Relate event to the user-visible workflow.
sequence_number	Stable ordering within session or run.
event_type / tool_name	profile, provider, bundle, scoring, validation, llm or system.
status	queued, running, completed, failed, skipped, cached.
started_at / completed_at / duration_ms	Operational timing.
cache_state	Live or cached classification displayed safely.
sanitized_arguments_jsonb	Approved non-secret argument summary.
user_safe_summary	Short UI progress message.
safe_error_code / safe_error_message	Sanitized failure information.

Chat Messages And Private Artifacts

Conversation data may reference trusted structured state, but prose never becomes scoring evidence.

Table 11 - chat_messages

Group	Columns	Rule
Identity	id; session_id; sequence_number; created_at	Unique(session_id, sequence_number).
Message	role; message_type; content	role: user, assistant, system_safe_summary, tool_status.
Trusted references	farm_profile_id; recommendation_run_id; validation_report_id	References structured state; prose itself is not trusted input.
Structured context	structured_context_jsonb	Selected crop, unresolved questions and safe IDs only; no chain-of-thought.
Generation	model_provider; model_name; generation_id; prompt_template_version; prompt_template_hash	Supports server-side model changes without coupling scores to the model.
Metrics	input_tokens; output_tokens; latency_ms; finish_reason	Optional cost and latency tracking.
Safety	safety_status; error_code; redaction_status	Store only user-safe content.

LLM BOUNDARY

The LLM may ask follow-up questions, select approved tools and explain validated results. It may not calculate suitability, alter confidence, remove risks, invent thresholds or authorize rendering.

Private Storage Bucket

Item	Contract
Bucket	provider-artifacts; server writes only; signed read URLs are short-lived and never stored.
Provider JSON	regions/{site_id}/{provider}/{request_hash}.json or locations/{location_hash}/{provider}/{request_hash}.json
FortyGuard map	heatmaps/{location_hash}/{activity_id}/{analytic_type}.geojson
Scenario exports	scenario-exports/{scenario_batch_id}/scenario_results.json and scenario_summary.csv
Presentation	presentations/{run_id}/{version}.json or .pdf when explicitly produced
Metadata in Postgres	bucket_name; object_path; sha256; content_type; size_bytes; provider/upload timestamps; sanitization and schema versions.

RLS, Immutability And Constraints

The browser uses the CropSage server API; all privileged operations remain server-side.

High-Value Constraints

Entity	Constraint
farm_profiles	Exactly one of planned_date/planned_month; valid Texas coordinate; version > 0.
provider_fetches	Reusable completed cache key uniqueness; no raw secret fields.
evidence_records	Valid unit; depth_top <= depth_bottom; appropriate value representation.
evidence_bundles	bundle_hash unique; completeness 0-100; validated requires snapshot and passed checks.
evidence_bundle_records	Composite PK(bundle_id, record_id).
recommendation_runs	Scenario requires parent_run_id; baseline forbids it; completed requires engine output.
crop_score_results	Scores 0-100; rank 1-22; unique(run_id,crop_id); unique non-null(run_id,rank).
validation_reports	Unique run; render_allowed=false when blocking errors exist.
run_events / chat_messages	Unique sequence number within owning session/run.

Required Indexes

Index	Purpose
GIST farm_profiles(farm_point / boundary)	Location and intersection queries.
provider_fetches(cache_key,status,expires_at)	Fresh-cache lookup.
provider_fetches(location_hash,provider)	Reuse evidence by location.
evidence_records(variable,valid_from,valid_to)	Retrieve time-aligned facts.
recommendation_runs(session_id,created_at desc)	Latest run and follow-up context.
crop_score_results(run_id,rank)	Ranked cards and comparisons.
run_events / chat_messages(session_id,sequence)	Ordered progress and conversation.

REQUIRED EXTENSIONS	Enable pgcrypto for gen_random_uuid()/hashing and postgis for farm geometry and spatial indexes.
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Migration Sequence And Launch Checks

A condensed implementation sequence that preserves the detailed schema and scoring contract.

Migration Sequence

#	Task	Acceptance check
1	Enable pgcrypto/PostGIS; create enums and helpers.	Extensions enabled; enum/timezone values tested.
2	Create assessment_sessions and farm_profiles with RLS/indexes.	Representative and arbitrary Texas profiles insert as immutable versions.
3	Create provider_fetches and private Storage policies.	Live, cached and failed request lifecycles are represented safely.
4	Create evidence_records, evidence_bundles and membership.	A known EvidenceBundle round-trips with identical hash.
5	Create recommendation_runs and crop_score_results.	A rank-all output persists exactly as 22 crop rows.
6	Create validation_reports and immutability triggers.	Invalid run cannot set render_allowed=true.
7	Create run_events, chat_messages and safe server policies.	Ordered progress/chat contains no secrets or hidden reasoning.
8	Implement cache lookup and Storage checksum verification.	Stable references and expiration behavior pass.
9	Seed 15 sites; run five scenarios each; freeze versions.	75 runs and 1,650 crop results reproduce with hashes.
10	Run RLS, provider failure, LLM failure and end-to-end tests.	Public demo renders validated output only.